

The first time we had our design and gathered the materials we measured the box wrong. We needed our platform to fit a 10x20 tray. We had made our original box out of a cabinet and we cut the pieces. Our first box was a 12x12 box. We made the box fine and it was stable but the measurements were wrong. Next we got the right measurements and made the box out of the wood that was not being used. This time the wood unfortunately did not work and was splitting. It was plywood so it was basically pieces of wood glued together. It would start to split whenever we tried to put a nail in it. After that we got better wood enough though the only wood that was available was bowed because it had been in the sun too long. This time we got the measurements right and were able to use this for our final design.

Attempt #1

- 1) Found cabinet that we took the wood from
- 2) Measured and cut the wood
- 3) Realized we did not measure the wood right

Attempt # 2

- 1) Got wood
- 2) Measured it
- 3) Tried to construct the box
- 4) Wood started to split when hammering in nails

Attempt #3

- 1) Got more reliable wood
- 2) Measured it to fit a 10x20 tray
- 3) Constructed it and made a door
- 4) Made pieces using the 3D printer to hold the blinds
- 5) Insert blinds and platform to hold the battery
- 6) Spraypaint and make final touches

Materials

- 1) Wood
- 2) Measuring tape
- 3) Circular saw
- 4) Electric drill
- 5) Hand saw
- 6) 3D printing machine
- 7) Spray paint
- 8) Sander
- 9) Jig saw
- 10) Nails
- 11) Hinges

Add ons to our design : We added a door to the side of the box so that we could easily slide in the tray for the plant. Designed wedges that hold the blinds. Added a battery to the stick on the blinds so it would turn them. We were going to add a timer so that they could turn at the same time every day. We spray painted the box black to make the design look more clean. We used

gears so the battery could spin to turn the blinds. We added a push button to make the blinds open and close based on a timer.

How would it allow us to collect data? It would've allowed us to collect data if we had finished earlier by putting a plant who maybe was overheated or burnt by the sun and trying to test out the timer. We could have checked on it and watered it everyday.

Attributes

- 1) Easy to use
- 2) Low maintenance
- 3) Environmentally friendly
- 4) Low cost

Details of our attributes

- 1) This design was very easy to use considering the fact that all you do is it outside or on a porch and the battery and timer do their job.
- 2) If the design works correctly it is not something that you have to worry about or check up on daily.
- 3) It doesn't require any gas or anything that would release chemicals that are bad for the environment.
- 4) The whole design and materials that we used only cost 18 dollars.

Testing the attributes

-Unfortunately our design had a malfunction the day before our designathon; where we would present our design. It had never had a malfunction until the day before. We stayed after school for 2 hours trying to fix it but unfortunately were not able to.